

Procedural Choice under Risk

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December 18, 2025 - TUS XI

Introduction

- Procedural choice is well-documented:
 - Kahneman & Tversky (1992): “When faced with a complex problem, people employ a variety of heuristic procedures in order to simplify the representation and the evaluation of prospects.”
- Procedural aspect of choice is important:
 - Herbert Simon (1978): “There is an urgent need to expand the established body of economics analysis... to encompass the procedural aspects of decision-making.”

Introduction

- Theoretical representation of procedural choice remains largely unexplored
 - Large amounts of potential procedures, difficult to organize
 - Procedural choice seen as incompatible with standard theories
 - Rubinstein 1988: “On the negative side, [this work] casts doubt as to whether human choices from among a set of lotteries are indeed transitive.”

This Paper

- This paper:
 - Focus on two fundamental procedures - **horizontal** and *vertical*
 - *Vertical*: evaluate lotteries by combining outcomes → pick the highest value
 - **Horizontal**: cancel out equivalent outcome pairs until one “dominates” the other

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 - Under regularity conditions, these procedures = EU.
 - *Vertical* relaxation: betweenness, rank-dependence, complexity aversion
 - **Horizontal** relaxation: skew-symmetric, regret

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 - Under regularity conditions, these procedures = EU.
 - *Vertical* relaxation: betweenness, rank-dependence, complexity aversion
 - **Horizontal** relaxation: skew-symmetric, regret
- Extend result to Savage

Disclaimer and Plan

- **Not** this paper:
 - A general definition of procedural choice
 - Claims of realism (as-if approach)

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- Plan today:
 - Vertical procedure
 - Horizontal procedure
 - Extension to Savage

Literature (Short)

- **Procedures:** Simon 1955, Rubinstein 1988, Arieli et. al. 2011, Halevy & Mayraz 2022, Arrieta & Nielsen 2025
- **Risk:** Dekel 1986, Chew 1989, Gul and Lantto 1990, Chateauneuf 1999, Cerreia-Vioglio, Dillenberger & Ortoleva 2015, Lanzani 2022, Puri 2023, He 2024, Mononen 2024
- Contributions:
 - Theoretical analysis of two fundamental procedures
 - Procedural interpretation of classical non-EU models

Combination

Procedures: Vertical

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Procedures: Vertical

- DMs often have to evaluate complex objects:
 - Ex.1 $1+34+11-7+21=?$
 - Ex.2 An insurance costing 200\$ per month which covers 70% of damages for any amount less than 1000\$ and 50% otherwise
- Common process across these: evaluating the whole by summing parts

Procedures: Vertical

Lottery ℓ_1

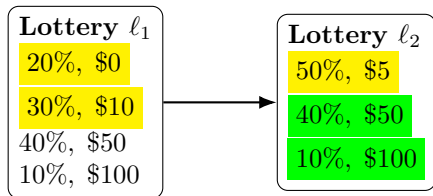
20%, \$0

30%, \$10

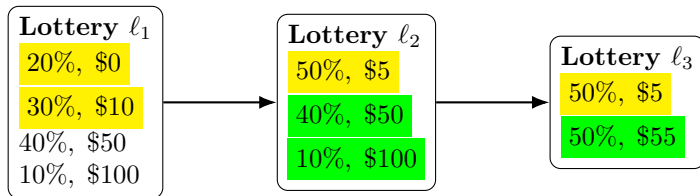
40%, \$50

10%, \$100

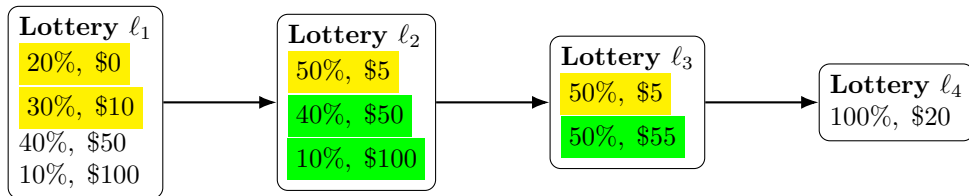
Procedures: Vertical



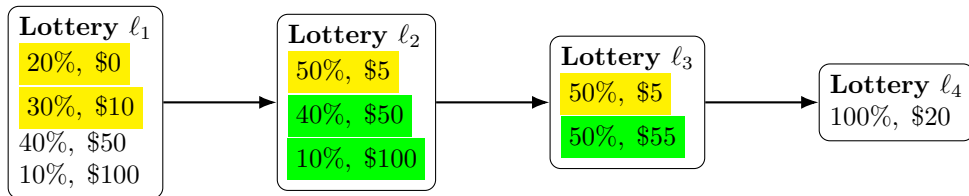
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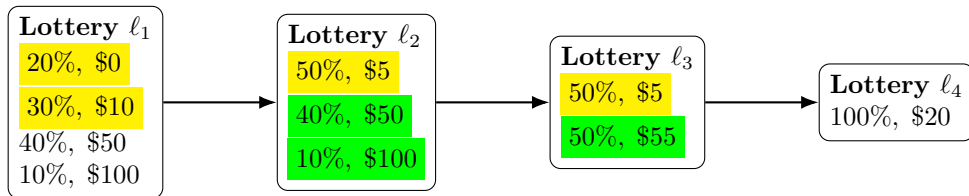


Procedures: Vertical



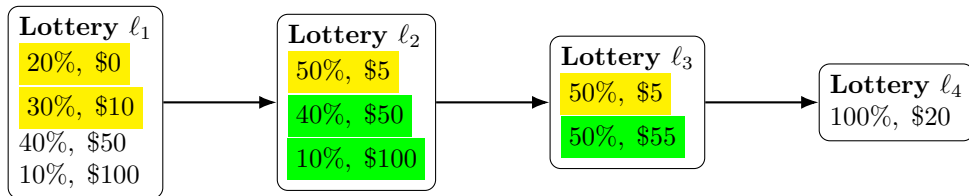
- The procedure: **Combination**

Procedures: Vertical



- The procedure: **Combination**
 - *Costless*: preserves indifference between lotteries

Procedures: Vertical



- The procedure: **Combination**
 - *Costless*: preserves indifference between lotteries
 - *Uniform*: combined outcome depends only on binary sublottery

Procedural Characterizations: Combination

Name	Axiom	Representation	Procedure
Expected Utility	Mixture Ind.	$V(\ell) = \sum_i p_i u(x_i)$	Uniform & Costless
Rank-Dependence	Comonotonic Ind.	$V(\ell) = \sum_{i=1}^n [u(x_i, \sum_{j=i}^n p_j) - u(x_i, \sum_{j=i+1}^n p_j)]$	Relax Uniformity
Betweenness	Betweenness	$V(\ell) = \sum_i p_i u(x_i, V(\ell))$	Relax Uniformity
Complexity Cost	Relax mon/cont.	$V(\ell) = \sum_i p_i u(x_i) - C(\ell)$	Relax Costlessness

Framework

- Monetary outcomes: $x \in [\underline{x}, \bar{x}]$
- Lotteries (finite support): $\ell \in \Delta(\mathcal{X})$
- Choice between lotteries: $\succsim$
- Let $a = (x_1, p_1; x_2, p_2)$ be a binary sublottery:
 - a is a binary sublottery of ℓ if $\exists \ell'$ such that $a + (1 - p_1 - p_2)\ell' = \ell$
 - $A = \{(a, \ell) \mid a \text{ is a binary sublottery of } \ell\}$

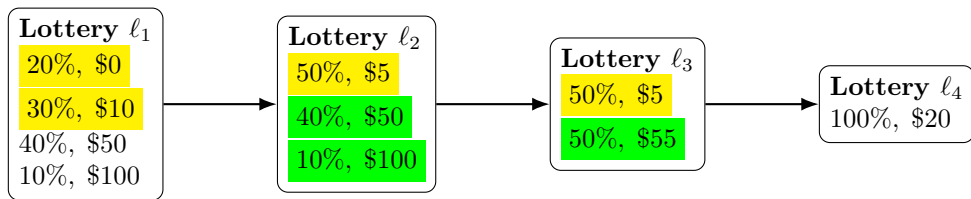
Combination

Definition

A **combination** rule is a mapping $I : A \rightarrow X$ such that

- ① **Monotone:** If ℓ and ℓ' differ only in a and a' , and a (strictly) fofd a' then $I(a, \ell)(\succ) \geq I(a', \ell')$
- ② **Continuous:** If $\lim \ell_n \rightarrow \ell$, then $\lim a_n \rightarrow a$ implies $\lim I(a_n, \ell_n) \rightarrow I(a, \ell)$
- ③ **Unique:** If ℓ becomes δ_x and $\delta_{x'}$ after sequences of combination then $x = x'$

The Procedure



- $I(\{(0, 0.2), (10, 0.3)\}, \ell_1) = 5$
- $I(\{(50, 0.4), (100, 0.1)\}, \ell_2) = 55$
- $I(\{(5, 0.5), (55, 0.5)\}, \ell_3) = 20$

Procedural Choice

Defined combination, but so far it has no choice implications - just a procedure

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Definition

$\succsim$ has a **uniform and costless procedural representation** if $\exists I$ such that:

- 1 Degenerate monotonicity: $x > y$ implies $\delta_x \succ \delta_y$
- 2 Costless combination: ℓ_1 becomes ℓ_2 after combination implies $\ell_1 \sim \ell_2$
- 3 Uniform combination: $I(a, \ell) = I(a, \ell')$ whenever a is a binary sublottery of ℓ and ℓ'

Interpretation: Compute “certainty equivalent” via combination and pick the largest

Expected Utility

Theorem

The following are equivalent:

- ① $\succsim$ has a uniform and costless procedural representation
- ② $\succsim$ has an expected utility representation with continuous and strictly increasing u

Proof: Straightforward application of Mixture Space Theorem [▶ Proof Sketch](#)

- Theorem 1: expected utility has a procedural representation, and we know exactly what kind of procedural regularities it satisfies.
- Natural question: what about relaxations of these regularities?
- This talk: relaxations of *uniformity*
 - ① Betweenness
 - ② Rank-Dependence
- Happy to show offline: complexity aversion via relaxing *costlessness*

Betweenness

Betweenness (Dekel 1986, Chew 1989)

- Recall *Mixture Independence*:

$$\ell \succeq \ell' \Rightarrow \forall \alpha, \ell^*, \alpha \ell + (1 - \alpha) \ell^* \succeq \alpha \ell' + (1 - \alpha) \ell^*$$

Betweenness (Dekel 1986, Chew 1989)

- Recall *Mixture Independence*:

$$\ell \succeq \ell' \Rightarrow \forall \alpha, \ell^*, \alpha \ell + (1 - \alpha) \ell^* \succsim \alpha \ell' + (1 - \alpha) \ell^*$$

- Natural and elegant weakening is *Betweenness*:

$$\ell \succeq \ell' \Rightarrow \forall \alpha, \ell \succsim \alpha \ell + (1 - \alpha) \ell' \succsim \ell'$$

Betweenness

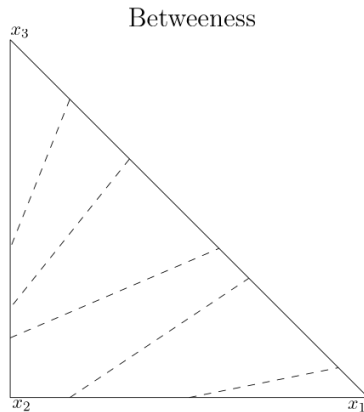
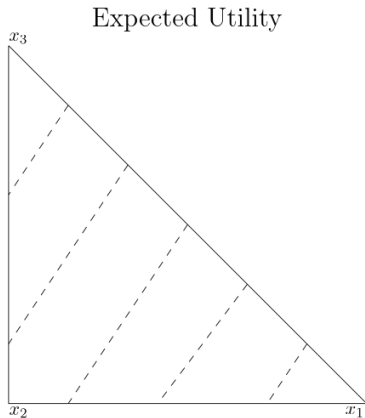
Theorem: Dekel 1986

The following are equivalent:

- ① $\succsim$ is a weak order, monotonic, continuous, and satisfies Betweenness
- ② There exists $u : [\underline{x}, \bar{x}] \times [0, 1] \rightarrow [0, 1]$ strictly increasing and continuous in the first argument and continuous in the second, where $\forall t \in [0, 1], u(\underline{x}, t) = 0$ and $u(\bar{x}, t) = 1$, such that $\succsim$ is represented by $V_{bt} : \mathcal{L} \rightarrow [0, 1]$ where $V_{bt}(\ell)$ is the unique number that solves:

$$V_{bt}(\ell) = \sum_{(x_i, p_i) \in \ell} p_i u(x_i, V_{bt}(\ell))$$

Betweenness Graphically



Linear but not necessarily parallel indifference curves

Sequential Uniformity

Definition: Uniformity

I is a uniform combination if $\forall \ell, \ell', I(a, \ell) = I(a, \ell')$

Sequential Uniformity

Definition: Uniformity

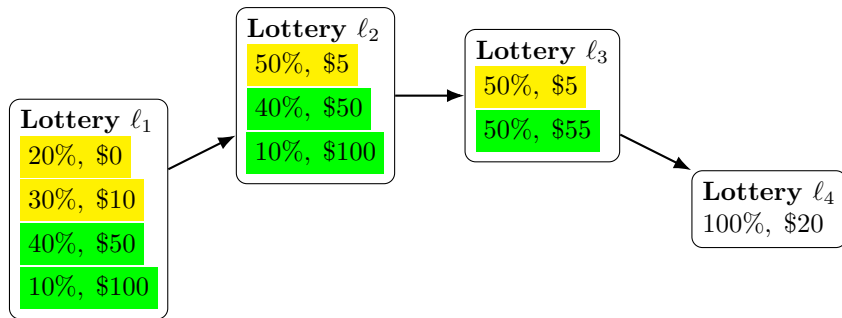
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Definition: Sequential Uniformity

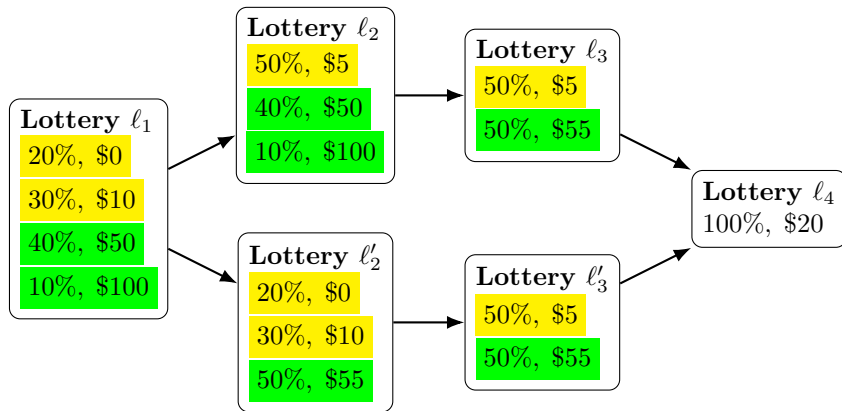
I is a **sequentially** uniform combination if $\forall \ell, \ell'$, if ℓ becomes ℓ' after an combination, then $I(a, \ell) = I(a, \ell')$

- Interpretation:
 - Uniformity with respect to order of combination
 - A form of perfect recall

Sequential Uniformity



Sequential Uniformity



Characterizing Betweenness

Theorem

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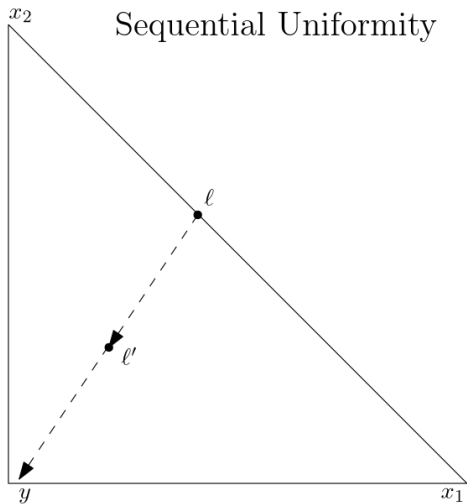
- ① $\succsim$ is such that there exists I and
 - ① $x > y$ implies $\delta_x \succ \delta_y$
 - ② I is costless
 - ③ I is sequentially uniform
- ② $\succsim$ has a Betweenness representation

► Proof Sketch

Intuition

- Consider the lottery $\ell = \{(x_1, p_1), (x_2, p_2)\}$ and let $a = \{(x_1, \frac{1}{2}p_1), (x_2, \frac{1}{2}p_2)\}$
- Sequential: if $I(a, \ell) = y$ and $\ell' = \{(x_1, \frac{1}{2}p_1), (x_2, \frac{1}{2}p_2), (y, \frac{1}{2}p_1 + \frac{1}{2}p_2)\}$ then $I(a, \ell') = y$
- By uniqueness, we know $\ell \sim \{(y, 1)\}$

Intuition



Rank-Dependence

Rank-Dependence

Definition: Sure-Thing Principle

$\ell_1 \succsim \ell_2$ and (x, p) is a sublottery of ℓ_1 and ℓ_2 , then can replace (x, p) by (y, p) in both lotteries without changing the preference

- Assume $x_n \geq x_{n-1} \dots \geq x_1$

Definition: Comonotonic Sure-Thing Principle

Let $\ell_1 = \{(x_1, p_1), \dots, (x_n, p_n)\}$ and $\ell_2 = \{(y_1, p_1), \dots, (y_n, p_n)\}$ be two lotteries distinct outcomes, then $x_k = y_k$ and $\ell_1 \succsim \ell_2$ implies that replacing x_k, y_k by some $x'_k = y'_k$ without changing the rank of outcomes preserves preference

Rank-Dependence (Chateauneuf 1999)

Theorem: Chateauneuf 1999

The following are equivalent:

- ① $\succsim$ is a weak order, monotonic, continuous, and satisfies the comonotonic sure-thing principle
- ② $\succsim$ has a rank-dependent ~~expected~~ utility representation, there exists $u : X \times [0, 1] \rightarrow \mathbb{R}$ strictly increasing with $u(x, 0) = 0$ and some additional regularities such that V_{rd} represents $\succsim$

$$V_{rd}(\ell) = \sum_{i=1}^n \left[u(x_i, \sum_{j=i}^n p_j) - u(x_i, \sum_{j=i+1}^n p_j) \right]$$

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- Many other axiomatizations of stronger/weaker forms of rank-dependence - all use the notion of rank within the axioms

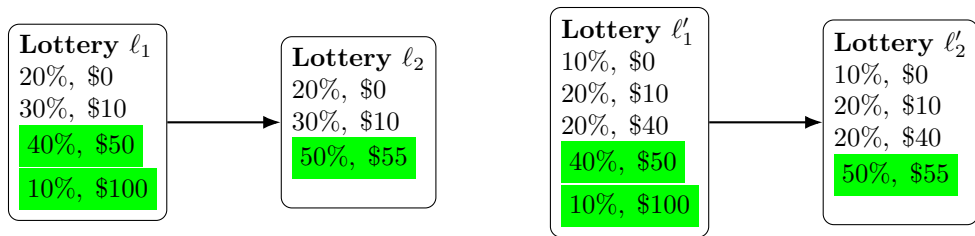
Maximal/Minimal Binary Sublotteries

- Procedurally, it turns out that rank-dependence can arise from a more primitive concern

Maximal/Minimal Binary Sublotteries

- Procedurally, it turns out that rank-dependence can arise from a more primitive concern
- A binary sublottery is a **minimal** (*maximal*) sublottery of ℓ if its outcomes are the two **smallest** (*largest*) in the support of ℓ
- A binary sublottery is an **extremal** sublottery of ℓ if its outcomes are each the largest and smallest outcomes.

Exterior Uniformity A

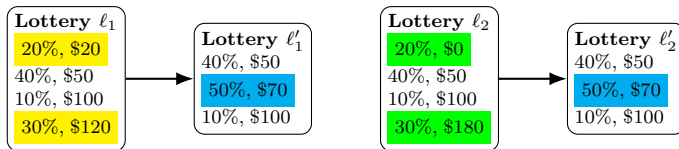


Definition: Exterior Uniformity A

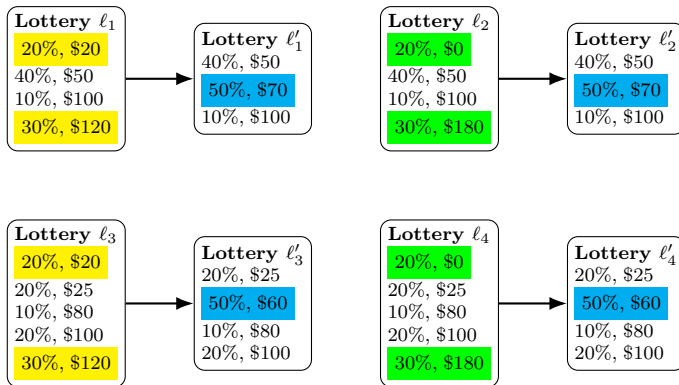
I satisfies exterior uniformity A if a is a **minimal** (*maximal*) sublottery of ℓ and ℓ' then $I(a, \ell) = I(a, \ell')$.

- DM orders outcomes and consistently combines maximal and minimal binary sublotteries across lotteries

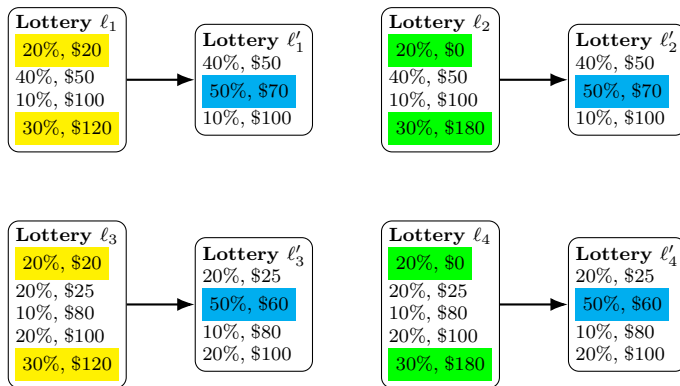
Exterior Uniformity B



Exterior Uniformity B



Exterior Uniformity B



Definition: Exterior Uniformity B

Suppose ℓ_1 and ℓ_2 differ only in extremal a and b . If $I(a, \ell_1) = I(b, \ell_2)$ then $I(a, \ell'_1) = I(b, \ell'_2)$ for any other ℓ'_1 and ℓ'_2 that differ on a and b extremally only.

Characterizing Rank-Dependent Utility

Theorem

The following are equivalent:

- ① $\succsim$ is such that there exists a combination I such that
 - ① $x > y$ implies $\delta_x \succ \delta_y$
 - ② I is costless
 - ③ I satisfies exterior uniformity A and B
- ② $\succsim$ has a rank-dependent utility representation

► Intuition

Rank-Dependent Expected Utility

Definition

$\succsim$ has a rank-dependent **expected** utility representation there exists $f : [0, 1] \rightarrow [0, 1]$ continuous, increasing, $f(0) = 0$, $f(1) = 1$ and $u : \mathcal{X} \rightarrow \mathbb{R}$ strictly increasing and continuous such that $\succsim$ is represented by V_{rdeu} :

$$V_{rdeu}(\ell) = \sum_{i=1}^n \left(f \left(\sum_{j=1}^n p_j \right) - f \left(\sum_{j=i+1}^n p_j \right) \right) u(x_i)$$

A more common and stronger form of rank-dependence. It requires another axiom, and a procedural characterization can also be found

Cancellation

Cancellation

- Combination: *separate* evaluation of lotteries and choosing the best one
- Many choices are driven by comparative aspects, i.e., Allais 2nd choice

Statewise Comparison

	ℓ_A	ℓ_B
30%	\$0	\$20
50%	\$50	\$40
20%	\$100	\$80

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Framework

- Monetary outcomes: $x \in [\underline{x}, \bar{x}]$
- Correlated lotteries (finite support): $\pi \in \Delta(\mathcal{X} \times \mathcal{X})$
- $\pi \in \Pi$: the first lottery is chosen
- Let $b = (x_1, y_1; x_2, y_2; p_1, p_2)$ be a pair of correlated outcomes
 - b is a pair correlated outcomes of π if $\exists \pi'$ such that $b + (1 - p_1 - p_2)\pi' = \pi$
 - $\mathcal{B}$ denotes the set of all pairs of correlated outcomes with $y_1, y_2 \in (x_2, x_1)$ or $x_1, x_2 \in (y_2, y_1)$

Cancellation

Definition

A **cancellation** rule $C \subseteq \mathcal{B}$ is a set of correlated pairs such that:

- **Symmetry:** $(x_1, y_1; x_2, y_2; p_1, p_2) \in C$ if and only if $(y_1, x_1; y_2, x_2; p_1, p_2) \in C$
- **Existence:** $\forall (x_1, y_1; x_2, y_2; p_1, p_2) \in \mathcal{B}$, one of the following holds
 - ① $(x_1, y_1; x_2, y_2; p_1, p_2) \in C$
 - ② $\exists p'_1 < p_1, (x_1, y_1; x_2, y_2; p'_1, p_2) \in C$
 - ③ $\exists p'_2 < p_2, (x_1, y_1; x_2, y_2; p_1, p'_2) \in C$
- **Monotone:** If $(x_1, y_1; x_2, y_2; p_1, p_2) \in C$ and $x'_1 > x_1$ then $(x'_1, y_1; x_2, y_2; p'_1, p_2)$ implies $p'_1 < p_1$.
- **Unique:** If π is such that one lottery strictly dominates another after a set of cancellation, then there does not exist another set with the opposite weak dominance.

Symmetry



Existence

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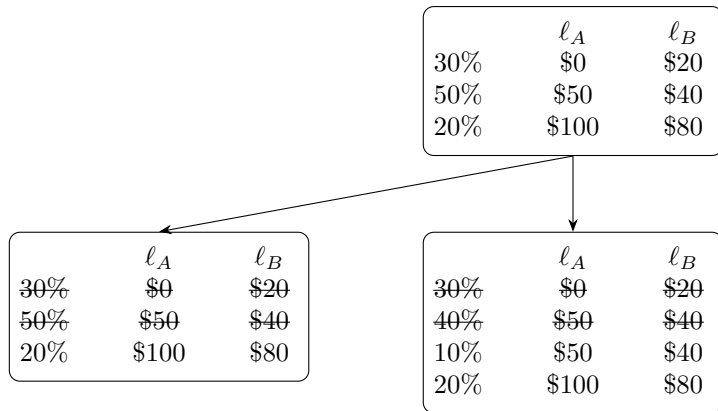
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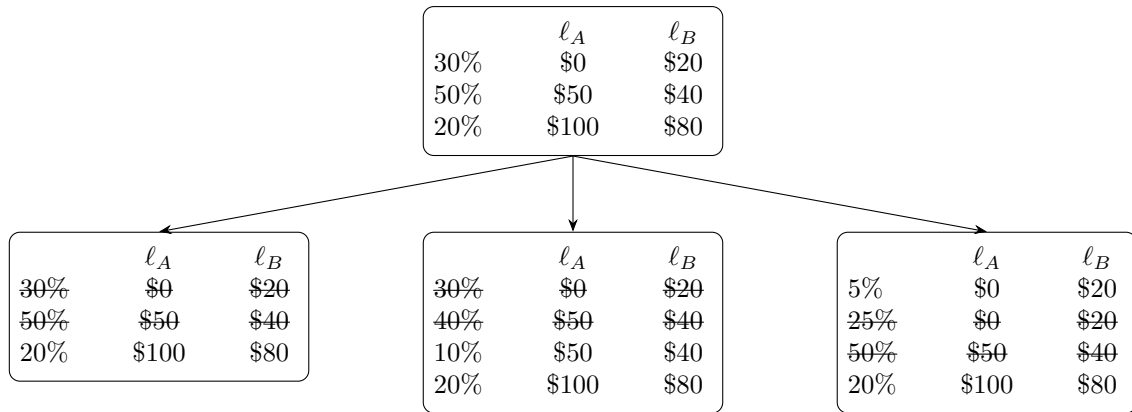


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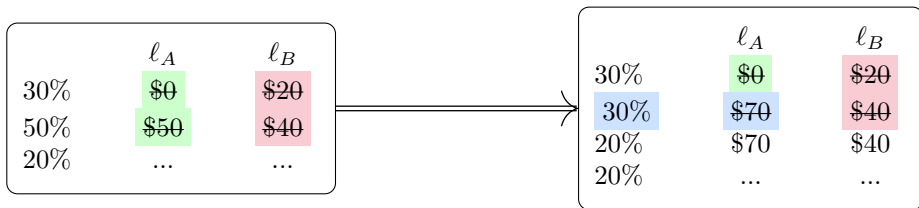
Existence



Existence



Monotonicity



Linking Procedure and Choice

Definition

Let Π be a preference on correlated lotteries. Π is consistent with a cancellation C when $\pi \in \Pi$ if and only if there exists a set of cancellations that leaves the first lottery of π to statewise dominate the second.

Representation

Theorem

The following are equivalent:

- ① *There exists a skew-symmetric $\phi : \mathcal{X}^2 \rightarrow \mathbb{R}$ that is strictly increasing and decreasing the first and second argument such that*

$$\pi \in \Pi \Leftrightarrow \sum_{(x,y)} \pi(x,y) \phi(x,y) \geq$$

- ② *Π is consistent with a cancellation C*

- Proof: verify Lanzani (2022) axioms.

Additional Properties

Definition: Additional Properties

- **Continuous:** If $\forall n, (x_1^n, y_1; x_2, y_2; p_1^n, p_2^n) \in C$ and $(x_1^n, p_1^n, p_2^n) \rightarrow (x_1, p_1, p_2)$ then $(x_1, y_1; x_2, y_2; p_1, p_2) \in C$

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- **Transitive:** If $(x_1, y_1; x_2, y_2; p_1, p_2) \in C$ and $(y_1, z_1; y_2, z_2; p_1, p_2) \in C$ then $(x_1, z_1; x_2, z_2; p_1, p_2) \in C$

Representation

Theorem

The following are equivalent:

- ① *There exists a skew symmetric and **continuous** $\phi : \mathcal{X}^2 \rightarrow \mathbb{R}$ that is strictly increasing and decreasing the first and second argument such that*

$$\pi \in \Pi \Leftrightarrow \sum_{(x,y)} \pi(x,y) \phi(x,y) \geq 0$$

- ② *Π is consistent with a cancellation C satisfying **continuity**.*

- Proof: verify Lanzani (2022) axioms.

Representation

Theorem

The following are equivalent:

- ① Π has an expected utility representation with strictly increasing and continuous u
- ② Π is consistent with a cancellation C satisfying symmetry, existence, monotonicity, uniqueness, continuity, and **transitivity**.
- ③ Π is consistent with a combination I satisfying monotonicity, continuity, uniqueness, uniformity, and costlessness.

- Proof: verify Lanzani (2022) axioms.

Definition

$\pi \in \Pi$ has a regret theory representation if there exists a continuous and increasing function u , a continuous, skew-symmetric ϕ that is strictly increasing and decreasing the first and second argument such that:

$$\pi \in \Pi \Leftrightarrow \sum_{(x,y)} \pi(x,y) \phi(u(x) - u(y)) \geq 0$$

Definition: Swapping Independence (He, 2024)

Swapping independent if $\forall (x_1, x_2; y_1, y_2; \frac{1}{2}, \frac{1}{2}) \in C$ then $(x_1, x_2; y_2, y_1; \frac{1}{2}, \frac{1}{2}) \in C$.

Swapping Independence

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50%	\$0	\$20
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Swapping Independence

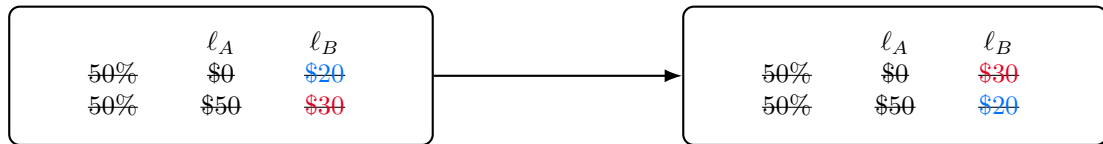
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Swapping Independence

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Swapping Independence



Regret Theory

Theorem

The following are equivalent:

- ① Π has a regret representation.
- ② Π is consistent with a cancellation C satisfying symmetry, uniqueness, existence, monotonicity, continuity, and swapping independence.

Extension to Savage

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- Straightforward to redefine procedures as combination/cancellation of events

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- Need additional regularity: event continuity, event/taste independence, probabilistic sophistication
- Extension technique: express the above procedurally $\rightarrow$ collapse back to risk
- Procedures do not depend on mixture operation - Savage is a natural framework

- $x \in [\underline{x}, \overline{x}]$ set of consequences
- $s \in \mathcal{S}$ set of states, $E \in \mathcal{E}$, set of events
- $f : S \rightarrow \mathcal{S}$, set of simple Savage acts

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- Binary sub-events: $\{(E_1, E_2, f) \mid E_1 \cap E_2 = \emptyset \text{ and } f \text{ single valued on } E_1, E_2\} = \mathcal{B}$
- Combination: $I : \mathcal{B} \rightarrow [\underline{x}, \bar{x}]$

Combination in Savage

Definition: Some Properties of Combinations

- Monotone: $f =_{E^c} g$, $E_1 \cup E_2 = E$, $f(E_1) = g(E_1)$, $f(E_2) \geq g(E_2)$, then $I(E_1, E_2, f) \geq I(E_1, E_2, g)$
- Continuous: $\forall x > y, \exists \{E_i\}$ mutually exclusive and $\cup_i E_i = \mathcal{S}$ such that $I(E_i^c, E_i, xE_i \underline{x}) > y$ and $x > I(E_i^c, E_i, yE_i \bar{x})$
- Uniqueness: Sequences combinations always lead to the same degenerate outcome
- Uniform: $f(E_1) = g(E_1)$, $f(E_2) = g(E_2)$, then $I(E_1, E_2, f) = I(E_1, E_2, g)$
- Comparative Prob: $x > y$, $x' > y'$ then $I(E_1, E_1^c, xE_1 y) \geq I(E_2, E_2^c, xE_2 y)$ if and only if $I(E_1, E_1^c, x'E_1 y') \geq I(E_2, E_2^c, x'E_2 y')$

Savage EU Procedurally

Theorem

The following are equivalent:

- 1 $\succsim$ is consistent with a combination rule that is monotone, continuous, unique, uniform, costless, and satisfies comparative probability.
- 2 $\succsim$ satisfies Savage P1-P6

Savage EU Procedurally

Theorem

The following are equivalent:

- ① $\succsim$ is consistent with a combination rule that is monotone, continuous, unique, uniform, costless, and satisfies comparative probability.
- ② $\succsim$ satisfies Savage P1-P6

- Comments:

- Finiteness is artificially restrictive, can augment act space and get P7 "for free"
- Can relax uniformity like under risk $\Rightarrow$ strengthen comparative prob. to get probabilistic sophistication

Conclusion

- **Message:** natural procedures of combining/cancelling outcomes yield new insights for classical models
 - Typical intuition: procedural choice $\neq$ standard axiomatic theories - not always true
 - Offer a procedural interpretation of axiomatic theories
 - Vertical and horizontal operations linked with theories - formalize a common intuition
 - Extension of characterization from risk to Savage via procedures
- Future work: experimental angle + ambiguity and procedures in Savage

Thank You.

Cautious Expected Utility

Negative Certainty Independence

- Recall *Mixture Independence*:

$$\ell \succeq \ell' \Rightarrow \forall \alpha, \ell^*, \alpha \ell + (1 - \alpha) \ell^* \succeq \alpha \ell' + (1 - \alpha) \ell^*$$

Negative Certainty Independence

- Recall *Mixture Independence*:

$$\ell \succeq \ell' \Rightarrow \forall \alpha, \ell^*, \alpha \ell + (1 - \alpha) \ell^* \succsim \alpha \ell' + (1 - \alpha) \ell^*$$

- A weakening that captures the intuition of the Allais Paradox is *Negative Certainty Independence*:

$$\ell \succeq \delta_x \Rightarrow \forall \alpha, \ell^*, \alpha \ell + (1 - \alpha) \ell^* \succsim \alpha \delta_x + (1 - \alpha) \ell^*$$

Cautious Expected Utility

Theorem: Cerreia-Violgio, Dillenberger, & Ortoleva 2015

Let $c(\ell, u)$ denote the certainty equivalent of ℓ according to the utility function u . The following are equivalent:

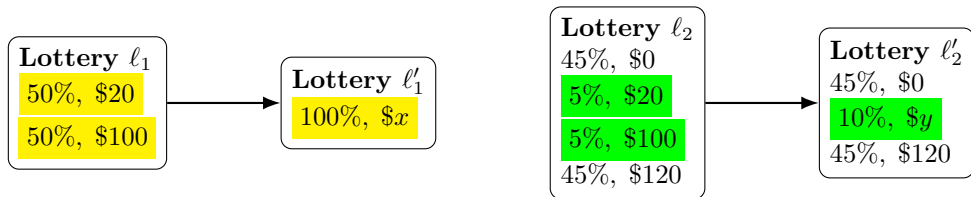
- $\succsim$ is a weak order, monotonic, continuous, and satisfies negative certainty independence
- There exists a set of continuous and strictly increasing utility functions $\mathcal{U}$ such that $\succsim$ is represented by V_{ceu}

$$V_{ceu}(\ell) = \inf_{u \in \mathcal{U}} c(\ell, u)$$

- Interpretation: uncertainty about own risk attitude, use the worst case

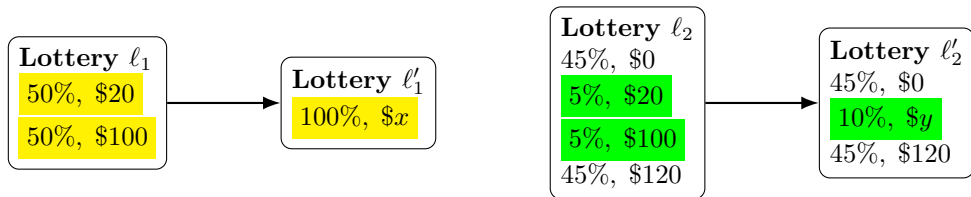
Procedural Intuition

- Consider merging the following two binary sublotteries:



Procedural Intuition

- Consider merging the following two binary sublotteries:



- We expect $x < y$ if the DM treats greater probabilities as carrying greater risk

Negative Proportional Sensitivity

- We say $\ell \approx \ell'$ if the lotteries are identical distributions, differ in visual representation only.

Definition: Negative Proportional Sensitivity (Informal)

I satisfies NPS if $\forall \ell, \alpha, \ell^*, \exists \ell' \approx \ell$ such there exists a sequence of combination of ℓ' to a degenerate outcome that “step-wise dominates” that of $\alpha \ell' \subseteq \alpha \ell + (1 - \alpha) \ell^*$.

- In words: ℓ can be rewritten such that a sequence of combination of ℓ is dominated by one of ℓ after mixing.
- Interpretation: Mixtures shrink the probabilities and therefore do not make all binary sublotteries combine to strictly worse outcomes

Procedural Representation

Theorem

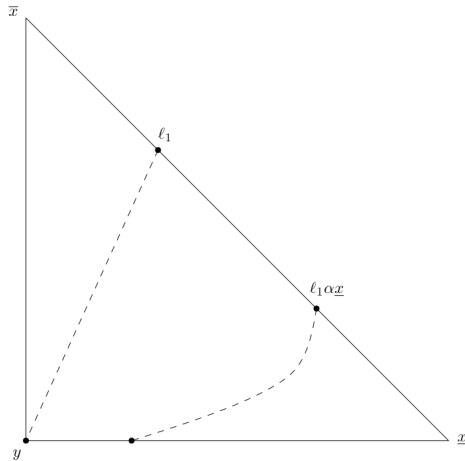
The following are equivalent:

- ① $\succsim$ is such that there exists an combination I such that
 - ① $x > y$ implies $\delta_x \succ \delta_y$
 - ② I is costless
 - ③ I satisfies negative proportional sensitivity
- ② $\succsim$ has a cautious expected utility representation

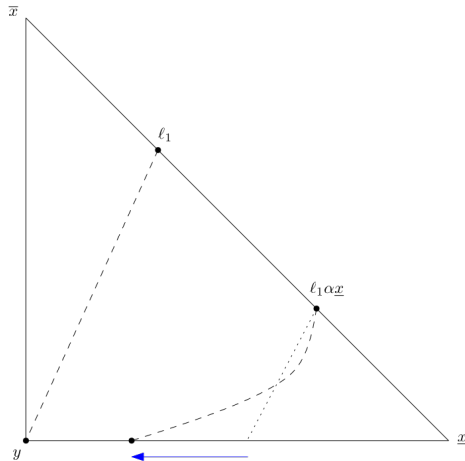
Intuition

- Consider the lottery $\ell = \{(x_1, p_1), (x_1, p_1), (x_2, p_2), (x_2, p_2)\}$ and let $a = \{(x_1, p_1), (x_2, p_2)\}$
- Sequential: if $I(a, \ell) = y$ and $\ell' = \{(x_1, p_1), (x_2, p_2), (y, p_1 + p_2)\}$ then $I(a, \ell') = y$
- By uniqueness, we know $\ell \sim \{(y, 1)\}$

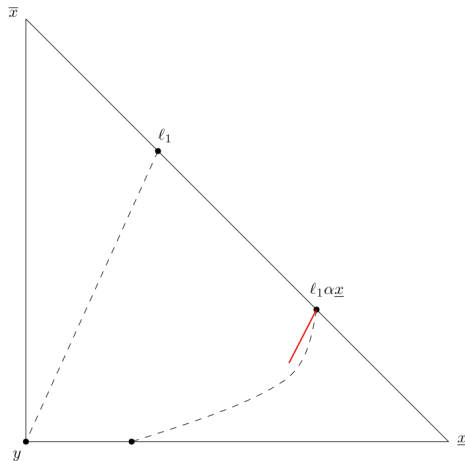
Intuition



Intuition



Intuition



Complexity Aversion (Briefly)

Complexity Aversion

- Two models of complexity aversion:

- Support-size cost: $V_{size}(\ell) = \sum_{(x_i, p_i) \in \ell} u(x_i)p_i - C(|\ell|)$

- Entropy cost: $V_{ent}(\ell) = \sum_{(x_i, p_i) \in \ell} u(x_i)p_i - \gamma H(\ell)$

Relaxation of Costlessness

- Costlessness implies that preference is preserved for *any* sequence of combinations.
 - $\ell_1 \succsim \ell_2$ then $\ell_1 \rightarrow \ell'_1$ and $\ell_2 \rightarrow \ell'_2$ by **any** two sequences of combinations implies $\ell'_1 \succsim \ell'_2$
- Natural weakening: preference preserved for pairs of **legal** sequences.

Relaxation of Costlessness

- Won't go into the details today, in general:
 - Counting legality: preserves preference if the preferred lottery's sequence of combination is shorter
 - Distributional legality: preserves preference if the two sequences merged binary sublotteries with same probability distributions

Relaxation of Costlessness

- Won't go into the details today, in general:
 - Counting legality: preserves preference if the preferred lottery's sequence of combination is shorter
 - Distributional legality: preserves preference if the two sequences merged binary sublotteries with same probability distributions
- With some additional details: $\text{Counting} \Leftrightarrow \text{Support Size} \wedge \text{Distributional} \Leftrightarrow \text{Entropy}$

Proof Sketch

- ① Costlessness + Mon/Cont/Uniq of $I \Rightarrow$ Well-Order/Mon/Cont of $\succsim$

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 - ② Can show uniformity implies $I(a, \ell) = I(\alpha a, \ell')$

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 - ③ $\alpha\ell_1 + (1 - \alpha)\ell^*$ becomes $x_1 + (1 - \alpha)x^*$ and $\alpha\ell_2 + (1 - \alpha)\ell^*$ becomes $\alpha x_2 + (1 - \alpha)x^*$

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 - ④ Monotonicity finishes the proof

◀ Back

Proof Sketch

- ① For every lottery ℓ , $\exists \alpha_\ell$ such that $\ell \sim \alpha_\ell \bar{x} + (1 - \alpha_\ell) \underline{x}$

Proof Sketch

- 1 For every lottery ℓ , $\exists \alpha_\ell$ such that $\ell \sim \alpha_\ell \bar{x} + (1 - \alpha_\ell) \underline{x}$
- 2 Can show there is a sequence of combination from $\alpha_\ell \bar{x} + (1 - \alpha_\ell) \underline{x}$ to ℓ (induction on cardinality of ℓ)

Proof Sketch

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- ② Can show there is a sequence of combination from $\alpha_\ell \bar{x} + (1 - \alpha_\ell) \underline{x}$ to ℓ (induction on cardinality of ℓ)
- ③ Take $\ell_1 \sim \ell_2$, want to show $\beta \ell_1 + (1 - \beta) \ell_2 \sim \ell_1$:
 - ① Can show $\alpha_{\ell_1} \bar{x} + (1 - \alpha_{\ell_1}) \underline{x}$ combines to $\beta \ell_1 + (1 - \beta) [\alpha_{\ell_1} \bar{x} + (1 - \alpha_{\ell_1}) \underline{x}]$
 - ② But since $\alpha_{\ell_1} = \alpha_{\ell_2}$ we can combine $\alpha_{\ell_1} \bar{x} + (1 - \alpha_{\ell_1}) \underline{x}$ to $\beta [\alpha_{\ell_1} \bar{x} + (1 - \alpha_{\ell_1}) \underline{x}] + (1 - \beta) \ell_2$.
 - ③ Sequential combination implies $\beta [\alpha_{\ell_1} \bar{x} + (1 - \alpha_{\ell_1}) \underline{x}] + (1 - \beta) \ell_2$ combines to $\beta \ell_1 + (1 - \beta) \ell_2$

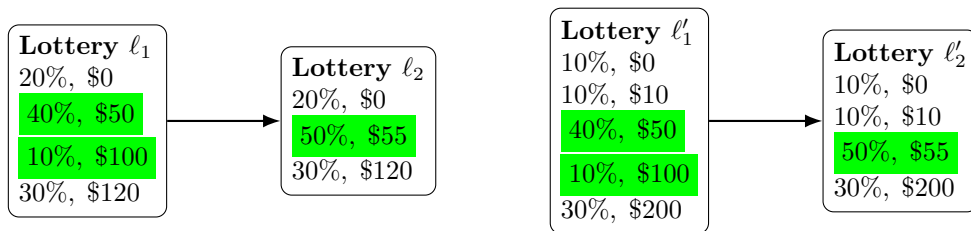
Proof Sketch

- ❶ For every lottery ℓ , $\exists \alpha_\ell$ such that $\ell \sim \alpha_\ell \bar{x} + (1 - \alpha_\ell) \underline{x}$
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- ❸ Take $\ell_1 \sim \ell_2$, want to show $\beta \ell_1 + (1 - \beta) \ell_2 \sim \ell_1$:
 - ❶ Can show $\alpha_{\ell_1} \bar{x} + (1 - \alpha_{\ell_1}) \underline{x}$ combines to $\beta \ell_1 + (1 - \beta) [\alpha_{\ell_1} \bar{x} + (1 - \alpha_{\ell_1}) \underline{x}]$
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 - ❸ Sequential combination implies $\beta [\alpha_{\ell_1} \bar{x} + (1 - \alpha_{\ell_1}) \underline{x}] + (1 - \beta) \ell_2$ combines to $\beta \ell_1 + (1 - \beta) \ell_2$
- ❹ Com/Tra/Cont/Mon given by regularity + costlessness

Rank-Dependence

[label=RankDependenceProof]Proof Sketch

- Exterior uniformity A implies already that combination of two *adjacent* outcomes depends only on the rank of these two outcomes



Rank-Dependence

Proof Sketch:

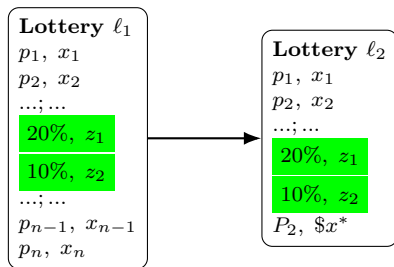
- ① Can show that integrating a min/max sublottery does not change the combination of other pairs

Rank-Dependence

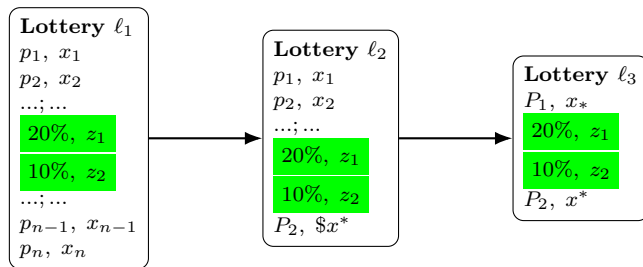
Proof Sketch:

- ① Can show that integrating a min/max sublottery does not change the combination of other pairs
- ② Then suffice to consider 4 outcome lotteries

Adjacent Lotteries



Adjacent Lotteries



Rank-Dependence

Proof Sketch (continued)

- Take $\ell_1 = \{(x_*, p_1), a, (x^*, p_2)\}$ and $\ell_2 = \{(y_*, p_1), a, (y^*, p_2)\}$. Suffice to show $I(a, \ell_1) = I(a, \ell_2)$

Rank-Dependence

Proof Sketch (continued)

- Take $\ell_1 = \{(x_*, p_1), a, (x^*, p_2)\}$ and $\ell_2 = \{(y_*, p_1), a, (y^*, p_2)\}$. Suffice to show $I(a, \ell_1) = I(a, \ell_2)$
- Let $\ell' = \{(x_*, p_1), a, (y^*, p_2)\}$

Rank-Dependence

Proof Sketch (continued)

- Take $\ell_1 = \{(x_*, p_1), a, (x^*, p_2)\}$ and $\ell_2 = \{(y_*, p_1), a, (y^*, p_2)\}$. Suffice to show $I(a, \ell_1) = I(a, \ell_2)$
- Let $\ell' = \{(x_*, p_1), a, (y^*, p_2)\}$
 - Then, $\{(x_*, p_1), a\}$ must be combined to the same outcome in ℓ_1 and ℓ' as we can find the same sequence of minimal sublotteries to combine it.

Rank-Dependence

Proof Sketch (continued)

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 - Suppose $I(a, \ell_1) \neq I(a, \ell')$, then combine a first, now by monotonicity + uniformity of minimal lottery we obtain $\{(x_*, p_1), a\}$ combines to two different outcomes in ℓ_1 and ℓ' .

Rank-Dependence

Proof Sketch (continued)

- Take $\ell_1 = \{(x_*, p_1), a, (x^*, p_2)\}$ and $\ell_2 = \{(y_*, p_1), a, (y^*, p_2)\}$. Suffice to show $I(a, \ell_1) = I(a, \ell_2)$
- Let $\ell' = \{(x_*, p_1), a, (y^*, p_2)\}$
 - Then, $\{(x_*, p_1), a\}$ must be combined to the same outcome in ℓ_1 and ℓ' as we can find the same sequence of minimal sublotteries to combine it.
 - Suppose $I(a, \ell_1) \neq I(a, \ell')$, then combine a first, now by monotonicity + uniformity of minimal lottery we obtain $\{(x_*, p_1), a\}$ combines to two different outcomes in ℓ_1 and ℓ' .
 - $I(a, \ell_1) = I(a, \ell')$ by contradiction. Same argument gives $I(a, \ell_2) = I(a, \ell')$